

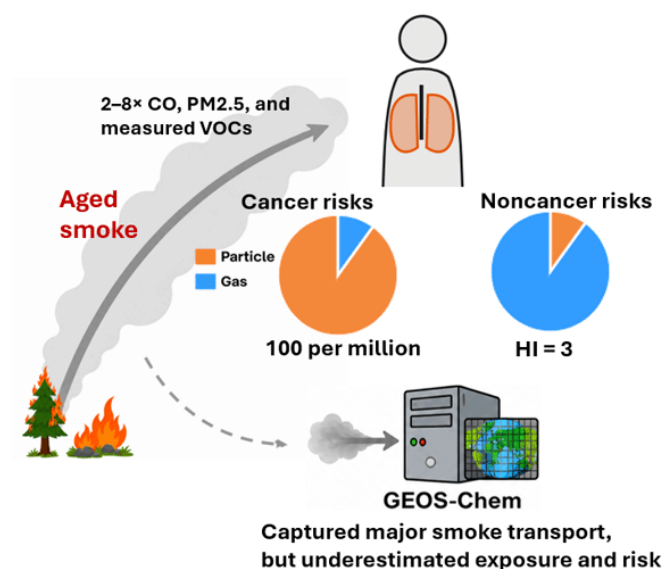
The Smoke Was Still There - but Its Chemistry Had Changed

Why more smoke did not always mean more ozone - and why the model underestimated exposure and risk

Jin et al. (2026) | Atmospheric Chemistry and Physics

In September 2020, a river of smoke crossed the western United States and reached the ground in Missoula, Montana. The main smoke mass had already spent at least three days traveling, mixing, and reacting in the atmosphere.

The team measured PM_{2.5}, carbon monoxide, nitrogen oxides, ozone, and 75 volatile organic compounds every hour, then tested the event with GEOS-Chem. Two contrasts defined the case: particles and organic gases surged while nitrogen oxides barely changed; ozone rose in lighter smoke but flattened or fell as the smoke thickened.



Key visual. Aged smoke raised ambient concentrations of particles, carbon monoxide, and organic gases. Particles and hazardous gases dominated different health-risk metrics, while GEOS-Chem captured the major transport but underestimated ambient concentrations and risk estimates.

Source: Jin et al. (2026), paper key visual, CC BY 4.0. <https://doi.org/10.5194/acp-26-11047-2026>

The questions behind the story: What had the smoke become after days in the atmosphere? Why did ozone stop rising as the plume thickened? And why could a model find the plume yet miss so much of what reached the air at ground level?

CHEMISTRY CONTINUED EN ROUTE

The plume traveled for days - and changed all the way

When the main smoke mass reached Missoula, carbon monoxide, $\text{PM}_{2.5}$, and the measured organic gases reached roughly 2–8 times the urban-background levels; hourly peaks reached 800 ppb, $120 \mu\text{g m}^{-3}$, and 85 ppb, respectively. Nitrogen oxides showed no clear increase. This was not fresh smoke with every ingredient scaled upward. It was a mixture rebuilt through dilution, oxidation, and secondary production.

Using carbon monoxide as a relatively stable dilution reference, the researchers followed VOC enhancement ratios across photochemical age. Here, photochemical age is an oxidation-based diagnostic, not exact travel time. Primary compounds such as benzene, toluene, and C8 aromatics declined as the plume aged. Methanol, formaldehyde, and acetaldehyde showed weaker or flatter trends, consistent with secondary production partly offsetting chemical loss; non-fire mixing may also contribute.

GEOS-Chem missed much of this chemical fingerprint. Its event-integrated enhancement ratio was about 35% low for benzene and 60%–80% low for the other key VOCs, while modeled OH exposure was about 100% too high. In the model, reactive gases were being consumed too quickly. Aged smoke was not merely diluted fresh smoke; it was a chemically rebuilt mixture.

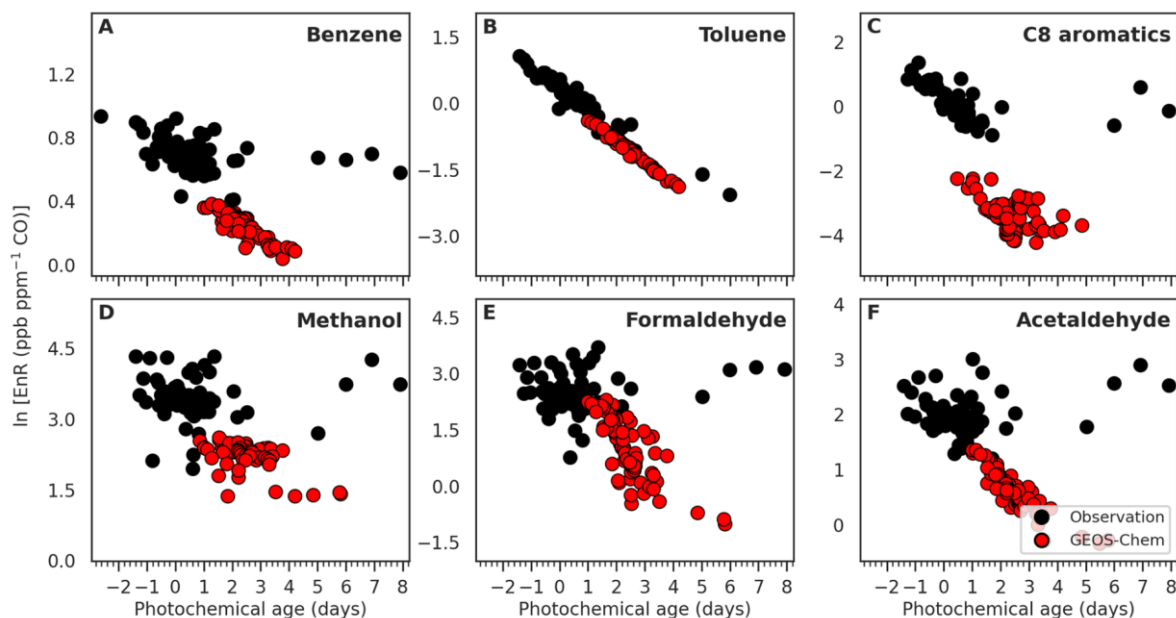


Figure 1. VOC enhancement ratios changed with photochemical age. Primary aromatics declined clearly; flatter oxygenated-VOC trends were consistent with secondary production offsetting part of the loss, although non-fire mixing may also contribute. GEOS-Chem generally underestimated the enhancement ratios.

Source: Jin et al. (2026), Fig. 5, CC BY 4.0.

A COUNTERINTUITIVE TURN

Why did ozone stop rising as the smoke became thicker?

Ozone is not emitted directly by a wildfire. It forms when organic gases and nitrogen oxides react in sunlight. Smoke can supply chemical ingredients while also blocking the light that drives the reaction. Without a concurrent increase in nitrogen oxides, the same plume may press both the accelerator and the brake on ozone production.

Observed maximum daily 8-hour ozone increased in lighter smoke. In this case study, which included a limited number of smoke days, ozone flattened or declined once $\text{PM}_{2.5}$ reached roughly $30\text{--}40\ \mu\text{g m}^{-3}$. During the strongest smoke, surface ozone fell by as much as about 15 ppb while incoming shortwave radiation dropped by around 50%. This was a descriptive turning point in one event, not a universal wildfire-smoke threshold.

GEOS-Chem and AIRPACT both continued to predict rising ozone as smoke loading increased. GEOS-Chem overestimated September MDA8 O_3 by about 15 ppb on average and by as much as roughly 70 ppb on smoke days. Reduced sunlight, the lack of a concurrent NO_x increase, a shallower boundary layer, and aged-plume chemical conditions may all have contributed; the study did not assign the error to one process alone.

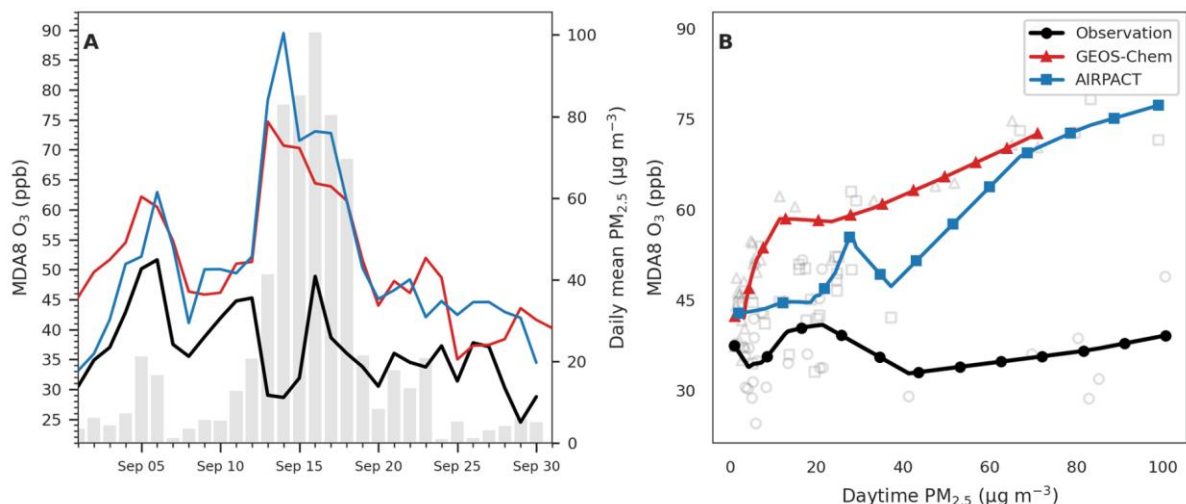


Figure 2. Observed ozone rose in lighter smoke but flattened or declined under heavier smoke. GEOS-Chem and AIRPACT continued to predict increasing ozone.

Source: Jin et al. (2026), Fig. 6, CC BY 4.0.

EXPOSURE BEYOND PARTICLES**Particles dominated one risk metric; hazardous gases dominated another**

The health analysis used an upper-bound screening scenario: a season as intense as 2020 was assumed to recur every year over a 70-year lifetime. The estimates compare long-term contributions from different pollutants. They are not observed cases from one smoke event and are not clinical predictions for an individual.

Under that scenario, smoke-attributable excess lifetime cancer risk was about 100 additional cases per million people, with roughly 90% attributed to PM_{2.5}. The 2020-style season increased the PM_{2.5}-attributable screening estimate by roughly sevenfold relative to the PM-related non-smoke baseline. Formaldehyde and benzene led the smaller gas-phase cancer contribution.

The chronic non-cancer hazard index was about 3, with about 88% attributed to hazardous gases and 12% to PM_{2.5}. Acrolein and formaldehyde were the leading gas-phase contributors. Particle filtration matters, but it cannot remove every hazardous gas; PM_{2.5} alone does not represent the full ambient exposure estimate.

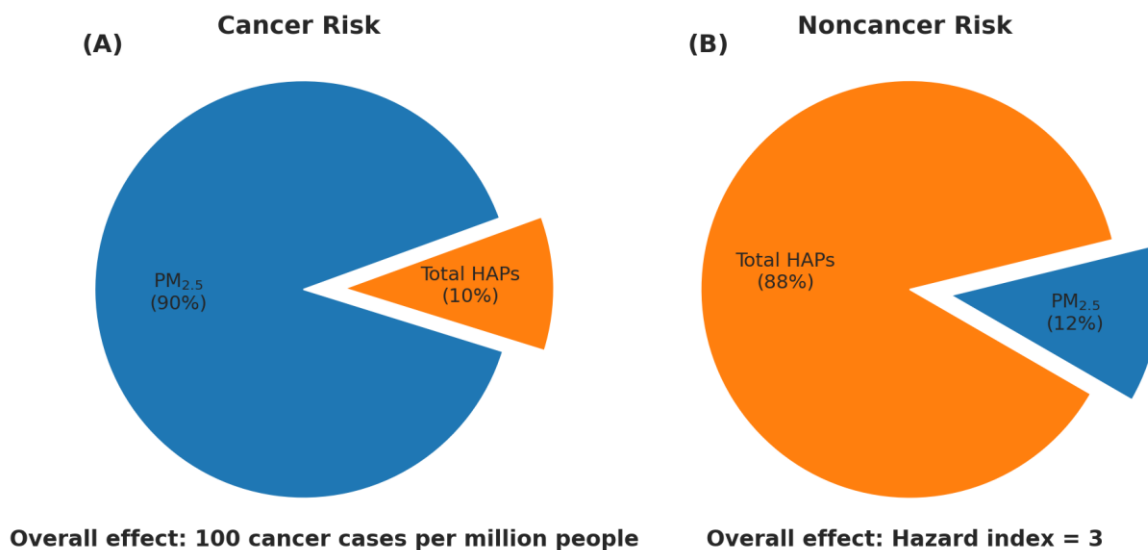


Figure 3. PM_{2.5} dominated the screening-level cancer estimate, while hazardous air pollutants dominated the chronic non-cancer hazard index.

Source: Jin et al. (2026), Fig. 7, CC BY 4.0.

How to read the numbers: The 100-per-million estimate belongs to the specified 70-year recurring-exposure scenario. A chronic hazard index of 3 signals the need for closer evaluation; it does not mean illness is certain, three times more likely, or clinically diagnosed.

THE MODEL FOUND THE PLUME BUT MISSED EXPOSURE

Capturing smoke transport was not the same as getting exposure right

For the first two episodes, baseline GEOS-Chem reproduced the hourly timing well ($R^2=0.8-0.9$), indicating that it largely captured when and where the smoke arrived. Yet it failed to reproduce the observed enhancements during the third episode and, during smoke periods, underestimated carbon monoxide and $PM_{2.5}$ by about 30%–40%, primary aromatics by 60%–80%, and most oxygenated VOCs by 50%–90%.

Tripling all biomass-burning emissions improved some primary gases and OH exposure, but shifted the $PM_{2.5}$ bias from about –30% to about +80%. The concentration errors then propagated into the risk estimates: the model produced 63 rather than 100 excess cancer cases per million and a chronic hazard index near 0.3 rather than 3. The issue was not one emissions multiplier; limitations in the model representation of smoke aging, secondary oxygenated-VOC production, and high-potency hazardous-air-pollutant chemistry also mattered.

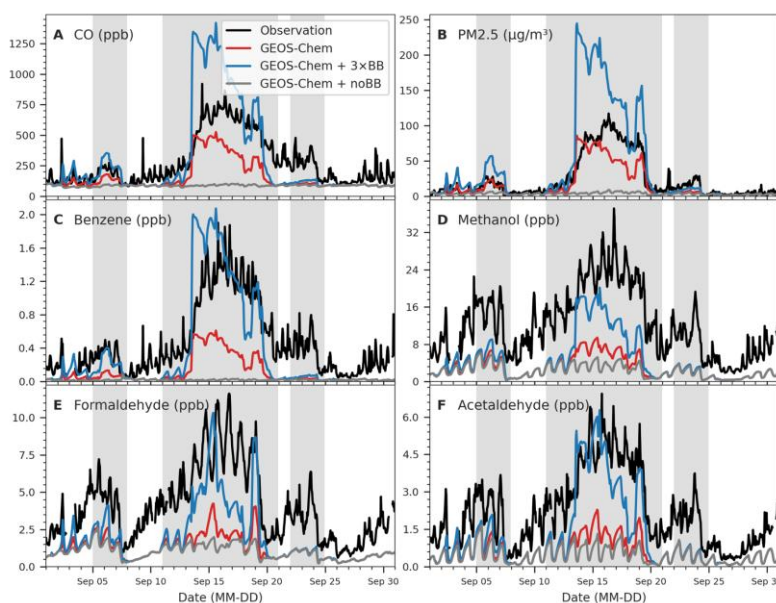


Figure 4. Observed concentrations compared with baseline GEOS-Chem, tripled biomass-burning emissions, and a biomass-burning-emissions-off (noBB) simulation. One emissions multiplier could not correct gases and particles at the same time.

Source: Jin et al. (2026), Fig. 8, CC BY 4.0.

The evidence chain: Emissions set the starting mixture. Days of chemistry reshape what reaches a city. Model errors can then propagate into exposure and risk estimates. “Where is the smoke?” is only the first forecasting question. The harder one is: “What is in it when it reaches the ground?”

Explore the study: Paper: [ACP article](#) | Data: [Zenodo](#) | Analysis code: [GitHub](#) | Archived code: [Zenodo](#)

This case study covers one city and one exceptional fire season; its numerical estimates should not be transferred directly to every fire or location. Figures and article content are available under CC BY 4.0. See the ACP paper and Supplement for methods and full context.